

Ethan Olsen

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Software engineer with full-stack, AI-product experience – evaluating and prompting frontier LLMs, building agentic workflows, and shipping data-driven applications end-to-end.

Background spans LLM evaluation, full-stack development (React, TypeScript, Python), database and data-pipeline management, and AI-assisted development, with experience owning products through the full development lifecycle across research, startup, and client environments.

Skills

Languages: Python, TypeScript/JavaScript, SQL, C/C++, Ruby, MATLAB.

AI / Agentic: LLM evaluation, agentic workflows, prompting strategies, context engineering, vector embeddings, semantic search, AI-assisted development (Claude Code, Codex, Antigravity).

Web / Backend: React, Next.js, Vite, FastAPI, PostgreSQL, Alembic.

Infrastructure / Tools: Docker, Linux/WSL, Git, pytest, Vitest, Playwright, Vercel, Railway, Render, Heroku, Supabase, Neon.

Work Experience

Surge AI | Remote USA

Research Fellow / Software Engineering | August 2023 - Present

- Research common failure modes of frontier LLMs in software engineering tasks.
- Designed 100+ adversarial conversations to induce frontier-LLM failures across 5+ evaluation dimensions, including safety, accuracy, code quality, honesty, and interaction quality.
- Documented 100+ LLM failure analyses through grading rubrics, behavioral analyses, and automated testing scripts using pytest, Vitest, and Playwright.
- Created 20+ containerized, network-isolated environments for evaluating LLM capabilities using Docker and Linux/WSL shell tooling.
- Solved software engineering tasks to establish reference standards, generating 100+ golden solutions as structured training data for frontier AI models – represented as repository diffs, validated repository states, and other reference artifacts across databases, backend systems, and full-stack applications.

Laplacent | Remote USA

Founder | September 2025 - Present

- <https://laplacent.com/>
- Built a pre-launch AI career platform combining semantic vector search, agentic workflows, and browser agents to automate job discovery and application workflows, owning the product through the full development lifecycle as sole engineer, from ideation and design through implementation, testing, and launch.
- Manage database schema design and data migrations using Alembic and PostgreSQL (Neon) to support the platform's semantic search and job-matching data pipeline.

- Delivered custom automation software for 2 small-business clients, developing end-to-end agentic workflows around internal business processes.

CadenceSEO | Remote USA

Software Engineer | March 2026 - July 2026

- Developed a page optimization platform that automated 200+ SEO optimization tasks per week, replacing roughly 60 hours of manual work per week, owning the product through the full development lifecycle as sole engineer. Scraped client web pages, evaluated keyword difficulty and content relevance, and recommended updates to titles, headers, and meta descriptions.
- Explored various prompting strategies to optimize the user-facing agentic workflows powering content-topic generation and page-optimization recommendations, improving output quality and relevance.
- Developed a content-topic generation platform that analyzed client keyword research and content coverage to identify 10+ keyword gaps per client. Deployed across 30+ clients, generating 300+ content topics per week.
- Managed database schema design and data migrations using Supabase (PostgreSQL) to support data for 30+ clients.

Arizona State University | Tempe, AZ

Research Intern / Advanced Semiconductor Packaging | June 2026 - July 2026

- Built and ran Ansys HFSS electromagnetic simulations to characterize signal integrity in advanced packaging structures, applying transmission line fundamentals (S-parameters, wave propagation, impedance matching) to interpret results.
- Gained cleanroom process experience (lithography, etching, CVD) at ASU NanoFab and MacroTechnology Works; performed die wire-bond testing on the Sigma Bond Tester.

Projects and Research

IR-Drop Prediction ML Model / Semiconductor Solutions Challenge | April 2026

Certificate: <https://credid.asu.edu/public/awards/dd47b4e7-db6e-4c96-96e1-61d30329e5b6X>

Github: https://github.com/ejom/Semiconductor-Solutions-Challenge-Submission_Ultron

- Designed a PyTorch U-Net fully convolutional model to predict power-delivery-network voltage drop across semiconductor layouts, pretrained on hundreds of synthetic circuit layouts and fine-tuned on real-circuit samples.
- Achieved RMSE = 0.54 mV, MAE = 0.45 mV, and NRMSE = 0.118 on held-out real-circuit data, with F1 = 0.510 for detecting high IR-drop regions (top 10% of ground-truth voltage-drop values).

Education

Northern Arizona University (GPA 3.78) | Flagstaff, AZ

M.S. Electrical Engineering | Expected December 2027 B.S. Electrical Engineering | Expected May 2027 GPA: 3.78

Relevant Coursework: Digital Signal Processing, Semiconductor Devices, FPGA Design, Embedded Systems